

PageRank

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Reputation Systems

Reputation System

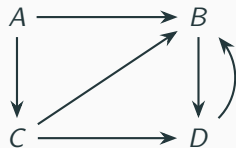
- Exists in many social networks from hotel reviews to search.
- Example 1: TripAdvisor allows users to rate hotels and restaurants and search for ones in a particular location.
- Example 2: Google uses a reputation system to rank web pages.

Naive PageRank

- Even the most obscure search can return a lot of results.
- Search engine must order the results in a way that is useful to the user.
- Example: “Used cars in New York City”.
- Page Rank is a method for ranking web pages.

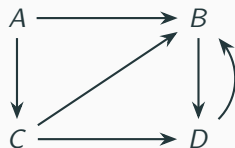
Naive PageRank

- If A links to B , then A is said to think highly of B .
- If many pages link to B , then B 's reputation is high.



Naive PageRank

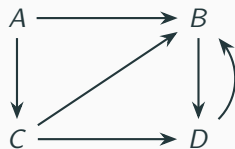
- Let d_v be the number of outgoing links from page v .
- Let π_v be the pagerank of page v , to be defined.
- Example: $d_C = 2$.



Naive PageRank

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- Example: $d_C = 2$.

$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u}$$



Definition

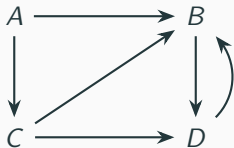
$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u}$$

- Constraint: $\sum_{v \in V} \pi_v = 1$ and $\pi_v \geq 0$ for all $v \in V$.
- Intuition: if u has an outgoing edge then u contributes $1/d_u$ to the pagerank of v .

Random Walk Interpretation

Random walk interpretation

- One can think of this as a monkey surfing the web.
- At each step, the monkey follows a link from the current page to the next page.



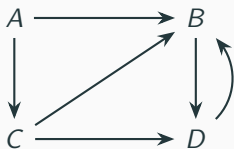
π_B can be interpreted as the fraction of time the monkey spends at B .

$$\pi_B = \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D.$$

Computing PageRank

PageRank's naive computation

Solve a system of linear equations.



$$\pi_A = 0$$

$$\pi_B = \frac{1}{2}\pi_A + \frac{1}{2}\pi_C + \pi_D$$

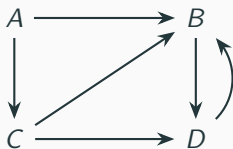
$$\pi_C = \frac{1}{2}\pi_A$$

$$\pi_D = \pi_B + \frac{1}{2}\pi_C$$

$$\pi_A + \pi_B + \pi_C + \pi_D = 1$$

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This gives $\pi_A = 0$, $\pi_B = \frac{1}{2}$, $\pi_C = 0$, $\pi_D = \frac{1}{2}$.

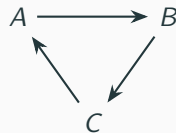
PageRank's naive computation

- The number of equations is the number of nodes n .
- $O(n^3)$ time using Gaussian elimination or matrix inversion.
- Infeasible for billions of web pages.
- We will deal with this later.

Flaws with Naive PageRank

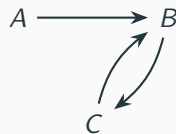
Flaws with naive PageRank (PageRank steal)

- A webpage cannot force other webpages to link to it.
- However, it can change its pagerank by changing its outgoing links.
- In the above, the pagerank of each page is $1/3$.
- Let see what happens if C changes its link to A to B .



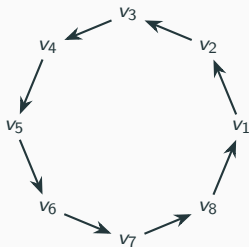
Flaws with naive PageRank (PageRank steal)

- Since A has no link to it, its pagerank is now 0.
- The pagerank of B and C is now $1/2$.
- Thus, C can “steal” pagerank from A .



Flaws with naive PageRank (PageRank steal)

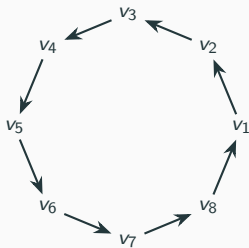
We can magnify the effect of this by creating a longer cycle.



All pageranks are equal which is $1/8$.

Flaws with naive PageRank (PageRank steal)

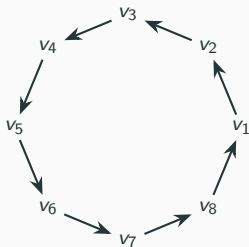
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Suppose v_2 redirects its link to v_3 toward v_1 . What happens?

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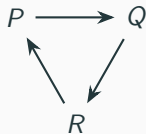
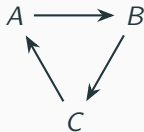
We can magnify the effect of this by creating a longer cycle.



Suppose v_2 redirects its link to v_3 toward v_1 . What happens? The new pageranks of $v_3, v_4, \dots, v_8$ are now 0. The new pagerank of v_1 and v_2 are now $1/2$.

Flaws with naive PageRank (Disconnected Components)

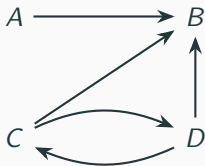
An even more serious flaw is that pagerank may not be well-defined.



- In the above, $\pi_A = \pi_B = \pi_C$ and $\pi_P = \pi_Q = \pi_R$.
- Recall:
$$\pi_A + \pi_B + \pi_C + \pi_P + \pi_Q + \pi_R = 1.$$
- $\pi_A = \pi_B = \pi_C = \pi_P = \pi_Q = \pi_R = 1/6$ is a solution.
- $\pi_A = \pi_B = \pi_C = 0$ and $\pi_P = \pi_Q = \pi_R = 1/3$ is another solution.

Flaws with naive PageRank (Dangling nodes)

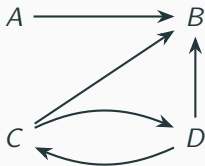
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- $\pi_A = 0$; $\pi_B = \pi_C/2 + \pi_D/2$;
 $\pi_C = \pi_D/2$; $\pi_D = \pi_C/2$.

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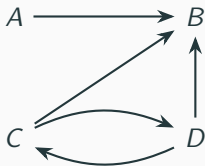
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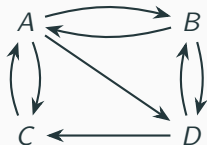
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- This implies that $\pi_C = \pi_D = 0$ and hence $\pi_B = 0$. But this violates the constraint $\pi_A + \pi_B + \pi_C + \pi_D = 1$.
- This means the solution might not even exist.

Matrix Formulation

Transition Matrix



The transition matrix with
 $A = 1, B = 2, C = 3, D = 4$:

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}.$$

$M_{ij} = 1/d_j$ if there is an edge from j to i , and
0 otherwise.

Power Iteration

Suppose the surfer is at a random page at time 0. Then $\pi^{(0)} = [1/4, 1/4, 1/4, 1/4]^T$.

The distribution at time 1 is $\pi^{(1)} = M\pi^{(0)}$:

$$\pi^{(1)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 3/8 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix} \approx \begin{bmatrix} 0.375 \\ 0.208 \\ 0.208 \\ 0.208 \end{bmatrix}.$$

Power Iteration (cont.)

Continuing to apply $\pi^{(t)} = M\pi^{(t-1)}$:

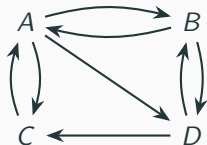
$$\pi^{(2)} = \begin{bmatrix} 15/48 \\ 11/48 \\ 11/48 \\ 11/48 \end{bmatrix} \approx \begin{bmatrix} 0.313 \\ 0.229 \\ 0.229 \\ 0.229 \end{bmatrix}$$

$$\pi^{(3)} = \begin{bmatrix} 11/32 \\ 7/32 \\ 7/32 \\ 7/32 \end{bmatrix} \approx \begin{bmatrix} 0.344 \\ 0.219 \\ 0.219 \\ 0.219 \end{bmatrix}$$

$$\pi^{(4)} = \begin{bmatrix} 63/192 \\ 43/192 \\ 43/192 \\ 43/192 \end{bmatrix} \approx \begin{bmatrix} 0.328 \\ 0.224 \\ 0.224 \\ 0.224 \end{bmatrix}$$

The distribution approaches a stationary π with $\pi = M\pi$. This gives the same definition as before.

Stationary Distribution



$M\pi = \pi$ gives:

$$\pi_A = \frac{1}{2}\pi_B + \pi_C$$

$$\pi_B = \frac{1}{3}\pi_A + \frac{1}{2}\pi_D$$

$$\pi_C = \frac{1}{3}\pi_A + \frac{1}{2}\pi_D$$

$$\pi_D = \frac{1}{3}\pi_A + \frac{1}{2}\pi_B$$

Equivalence of Definitions

The transition matrix encodes the same relation:

$$(M\pi)_i = \sum_j M_{ij} \pi_j = \sum_{j \rightarrow i} \frac{1}{d_j} \pi_j = \sum_{j \rightarrow i} \frac{\pi_j}{d_j} = \pi_i.$$

Thus the PageRank vector π is a stationary distribution ($\pi = M\pi$) of the random walk on the graph.

Conditions for Unique PageRank

The distribution approaches a stationary distribution π such that $\pi = M\pi$.

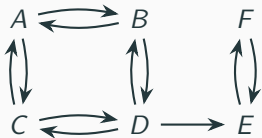
From Markov process theory, such π exists, is unique, and power iteration converges to it if:

- The graph is strongly connected (irreducible).
- The graph has no dangling nodes (dead ends).
- The graph is aperiodic: GCD of all cycle lengths is 1.

The ϵ -teleportation fix ensures all three hold simultaneously.

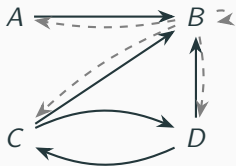
Fixing PageRank

Spider Traps



- The graph is strongly connected and has no dangling nodes.
- However, there is a spider trap: eventually, a random surfer will get stuck in E and F .
- The pagerank will be $1/2$ for E and F and 0 for the rest (which is not reasonable).

Fix 1: Dangling Nodes



- B is a sink: it has no outgoing edges.
- Fix: replace each sink with edges to *all* n nodes, each with weight $1/n$.
- Dashed edges show the added links from B .
- After the fix, every column of M sums to 1 (M is a proper stochastic matrix).

Fix 2: Teleportation

After handling sinks, there can still be spider traps and disconnected components.

Solution: allow the random surfer to jump to a page uniformly at random with probability ϵ (Google sets $\epsilon \approx 1/7$).

$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} + \epsilon \frac{1}{n}.$$

Revised Transition Matrix

The new transition matrix (e is the all-ones matrix):

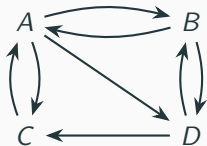
$$\hat{M} = (1 - \epsilon) M + \frac{\epsilon}{n} e.$$

$$\hat{M} = (1 - \epsilon) \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} + \epsilon \begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{bmatrix}.$$

To find the stationary distribution, solve $\pi = \hat{M}\pi$. In this case:

$$\pi_A = (1 - \epsilon)(\pi_B/2 + \pi_C) + \epsilon/4, \quad \pi_B = (1 - \epsilon)(\pi_A/3 + \pi_D/2) + \epsilon/4, \quad \dots$$

PageRank Example ($\epsilon = 0.12$)



With $\epsilon = 0.12$, $n = 4$, $\epsilon/n = 0.03$:

$$\pi_A = 0.88(\pi_B/2 + \pi_C) + 0.03$$

$$\pi_B = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_C = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_D = 0.88(\pi_A/3 + \pi_B/2) + 0.03$$

Solving: $\pi_A \approx 0.326$, $\pi_B \approx \pi_C \approx \pi_D \approx 0.224$.

Power Method

Computing PageRank: Power Method

- We compute PageRank iteratively (the power method).
- Start with $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$.
- For $t = 1, 2, \dots$, update $\pi^{(t)} = \hat{M}\pi^{(t-1)}$. Equivalently:

$$\pi_v^{(t)} = (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} + \epsilon \frac{1}{n}.$$

- The distribution $\pi^{(t)}$ converges to the stationary distribution π .
- Running time: $O(m + n)$ per iteration, where m is the number of edges.

Define the error at iteration t :

$$\text{error}(t) = \sum_v |\pi_v^{(t)} - \pi_v|.$$

Convergence Bound

$$\text{error}(t) \leq (1 - \epsilon)^t \cdot \text{error}(0).$$

- The error decreases by a factor of $(1 - \epsilon)$ each iteration.

Adjusted Computation: Stationary Equation

With the fix, a dangling node u contributes π_u/n to *every* page, while a normal node u contributes π_u/d_u to each neighbor.

The stationary equation $\pi = \hat{M}\pi$ becomes:

$$\pi_v = (1 - \epsilon) \left(\sum_{\substack{u \rightarrow v \\ d_u > 0}} \frac{\pi_u}{d_u} + \underbrace{\frac{1}{n} \sum_{\substack{u \\ d_u = 0}} \pi_u}_{\text{dangling mass}} \right) + \epsilon \frac{1}{n}.$$

Adjusted Computation: Power Iteration

The power-iteration update ($\pi^{(t)} = \hat{M}\pi^{(t-1)}$):

$$\pi_v^{(t)} = (1 - \epsilon) \left(\sum_{\substack{u \rightarrow v \\ d_u > 0}} \frac{\pi_u^{(t-1)}}{d_u} + \frac{1}{n} \sum_{\substack{u \\ d_u = 0}} \pi_u^{(t-1)} \right) + \epsilon \frac{1}{n}.$$

The dangling sum $\sum_{u: d_u=0} \pi_u^{(t-1)}$ is computed once per iteration, then added to every v .